Testing ANN and MPS Signals in FinRL

Short technical note | Aarav Nagar | July 2026

Question

My earlier project found that a quantum-inspired model could perform slightly better than an ANN on some stock-prediction metrics. Professor Liu suggested testing whether that difference still mattered when the prediction became part of an autonomous trading agent.

I asked whether an MPS prediction signal improves FinRL trading performance when the data, model size, PPO setup, transaction costs, and random seeds are held constant.

Setup

I used 15 stocks from the processed Nasdaq dataset used in the FinRL Contest 2025 workflow. The prediction models were fit on 2013-2017 data and validated on 2018. PPO was trained on 2013-2018 and tested out of sample on 2019-2023.

Condition	State given to PPO	Size
Base FinRL	Cash, prices, holdings, and 10 indicators per stock	181
ANN signal	Base state plus one ANN prediction per stock	196
MPS signal	Base state plus one MPS prediction per stock	196

The base state contains 1 cash value, 15 prices, 15 holdings, and 10 indicators for each stock. For the signal agents, I appended 15 frozen next-day return predictions as one additional indicator block. The prediction model was not updated during PPO training.

Both models used the same 13 inputs and had 369 trainable parameters. The ANN used 13-16-8-1 fully connected layers. The MPS used a trigonometric feature map with bond dimension 4. It ran on classical hardware and did not use qubits.

Each PPO condition used seeds 0, 1, and 2, a 5,000-step training budget, a $1,000,000 starting portfolio, and a 0.10% fee on every buy and sell. Turnover was calculated from executed traded notional divided by portfolio value at the start of the step.

Results

The MPS had slightly lower prediction error and higher directional accuracy, although its information coefficient was slightly negative.

Model	MSE	Directional accuracy	Information coefficient
ANN	1.5101	50.25%	+0.0119
MPS	1.5034	51.06%	-0.0060

The small prediction improvement did not carry over to trading.

Agent	Total return	Sharpe	Max drawdown	Turnover	Total cost
Base FinRL	73.24%	0.559	-40.46%	26.43%	$1,375.38
ANN signal	104.66%	0.735	-27.55%	27.32%	$1,429.65
MPS signal	94.27%	0.694	-26.79%	24.14%	$1,253.61
Equal-weight	218.25%	1.115	-28.54%	20.06%	$1,000.00

The PPO values are means across three seeds. MPS had a lower Sharpe than ANN in every paired run: 0.759 versus 0.790 for seed 0, 0.726 versus 0.812 for seed 1, and 0.596 versus 0.602 for seed 2.

Using a 20-trading-day block bootstrap with 2,000 samples, the MPS-minus-ANN annualized return difference was -0.92 percentage points. Its 95% interval was -4.63 to +2.63 points. Because that interval includes zero, this run does not show a reliable difference between the two signal agents.

MPS beat ANN in 2020 and 2023, while ANN was better in 2019, 2021, and 2022. Neither signal agent beat the equal-weight benchmark over the full test period.

What I learned

In this setup, the MPS signal did not improve trading performance over the equally sized ANN signal. It did slightly better on two prediction metrics, but the ANN produced a higher Sharpe in all three trading runs. This is the prediction-decision gap that motivated the study.

The result is limited to this experiment. I used one time split, three PPO seeds, one MPS bond dimension, and a relatively short 5,000-step training budget. The universe is also a 15-stock Nasdaq subset. The cost model includes a fixed transaction fee but not spread, slippage, market impact, or taxes.

Trading activity became very low later in the test period, suggesting that some policies settled into nearly static positions. Before testing a tensor-network policy architecture, I would repeat the signal experiment with more seeds, longer PPO training, and rolling retraining periods.

Reproduction

The repository includes the full pipeline, tests, daily equity curves, per-seed and yearly metrics, bootstrap output, and a run manifest with the pinned data checksum and FinRL commit.

Repository: https://github.com/Aarav-Nagar/QINN-FinRL-Evaluation

References

1. Biamonte, J. et al. “Quantum Machine Learning.” Nature 549, 195-202 (2017).
2. Liu, X.-Y. and Fang, Y. “Quantum Tensor Networks for Variational Reinforcement Learning.” NeurIPS 2020 workshop.
3. FinRL Contest 2025, Task 1: FinRL-DeepSeek for Stock Trading.
4. FinRL commit used here: 2334a5fe6d30629157f13c3b0319e1637e15e123.